

Theme Report 3

Act

COMPENG 2DX3

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Theme

The theme of this report is Act - in which it includes lab exercises 7-8. Acting is one of three fundamental properties of an intelligent micro system - centered around how such a system can interact with the world around it. This theme is demonstrated through real-world technologies, such as Autonomous Emergency Braking (AEB) in vehicles, which prevents or lessens severity of accidents that may occur on the road.

Background

Acting is important for intelligent microsystems to perform/execute actions mechanically and/or digitally. Without acting, there is no impact done by the microsystem to the environment even if the system has observed and reasoned information around it. This allows a purpose for the system to exist and execute the functions a programmer/designer has implemented for it to do. Modern systems require systems to act continuously to perform and serve user experience efficiently and effectively.

Lab 7 exercises required the implementation of interrupts through coding an interrupt service routine. It also required verification of data transmission between the MCU and ToF using both the Protocol and Logic workspace of the AD3. Lab 8 exercises required implementation of serial UART communication to transmit ToF sensor model ID and type. It also required motor movement and recording of data through the ToF sensor. These labs demonstrated how a microcontroller executes various actions corresponding to the reasoning it performs after the information it receives.

Theme Exemplars

Lab 7 - Interrupts and Event Programming

The first example selected from my lab work in relation to the theme - Act - is Milestone 2 Exercise from Lab 7. This exercise required event based programming; triggering a GPIO interrupt when User Switch 1 (PJ0) was pressed. Data from ISR is then transmitted to the ToF device, which is then verified using Protocol and Logic workspace of the AD3. This demonstrates the 'Act' portion of what the microsystem goes through - first Observing if an interrupt occurred, Reasoning if something happened to trigger an action, then Acting based on if something *did* happen. In this particular milestone, the microcontroller would act upon an interrupt by transmitting data to the ToF, verified using a 'Spy on the Line' - which is the AD3. This is demonstrated in *Figure 1*, in which we can see h5C (the data sent) labelled after the address (h29) - h standing for hex.

Some troubleshooting was also done for this lab portion, as the HI and LO signals were not showing up even after opening up the Logic from the AD3. It turned out that the rectangular button from the intersection of SDA and T had to be set to the falling edge. This is because it indicates when the AD3 should begin recording as the SDA is always high when using pull-up resistors.

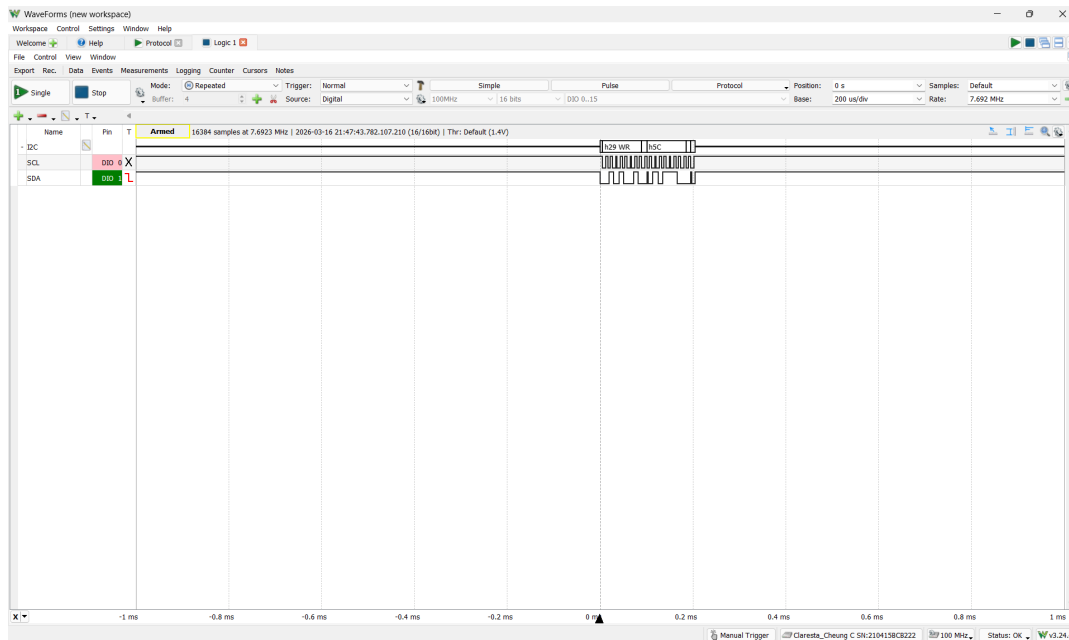


Figure 1: Spy on the Line

Lab 8 - Collecting Distance Data

The second example selected from my lab work in relation to the theme - Act - is Milestone 1 Exercise from Lab 8. This exercise required the implementation of serial UART communication to transmit the VL53L1X ToF sensor model ID and module type. Function calls were implemented as well to receive the required information from the ToF sensor in the MCU via I2C. This is then displayed on Realterm (*Figure 2*).

During the completion of this lab, I felt much frustration because the correct values were not displayed on RealTerm. I expected to see the model ID and type through hex numbers, but neither were displaying correctly. I only received 0x00, which was not expected. I read and re-read my code, wired and re-wired my connections. This was then troubleshooted by attaching

vi5311_platform instead of vi5311_platform_2dx4 in the files. It turned out to not be a hardware OR a code issue, but an airheaded decision of implementing the wrong files. The correct files along with some of the code are shown in *Figure 3*.

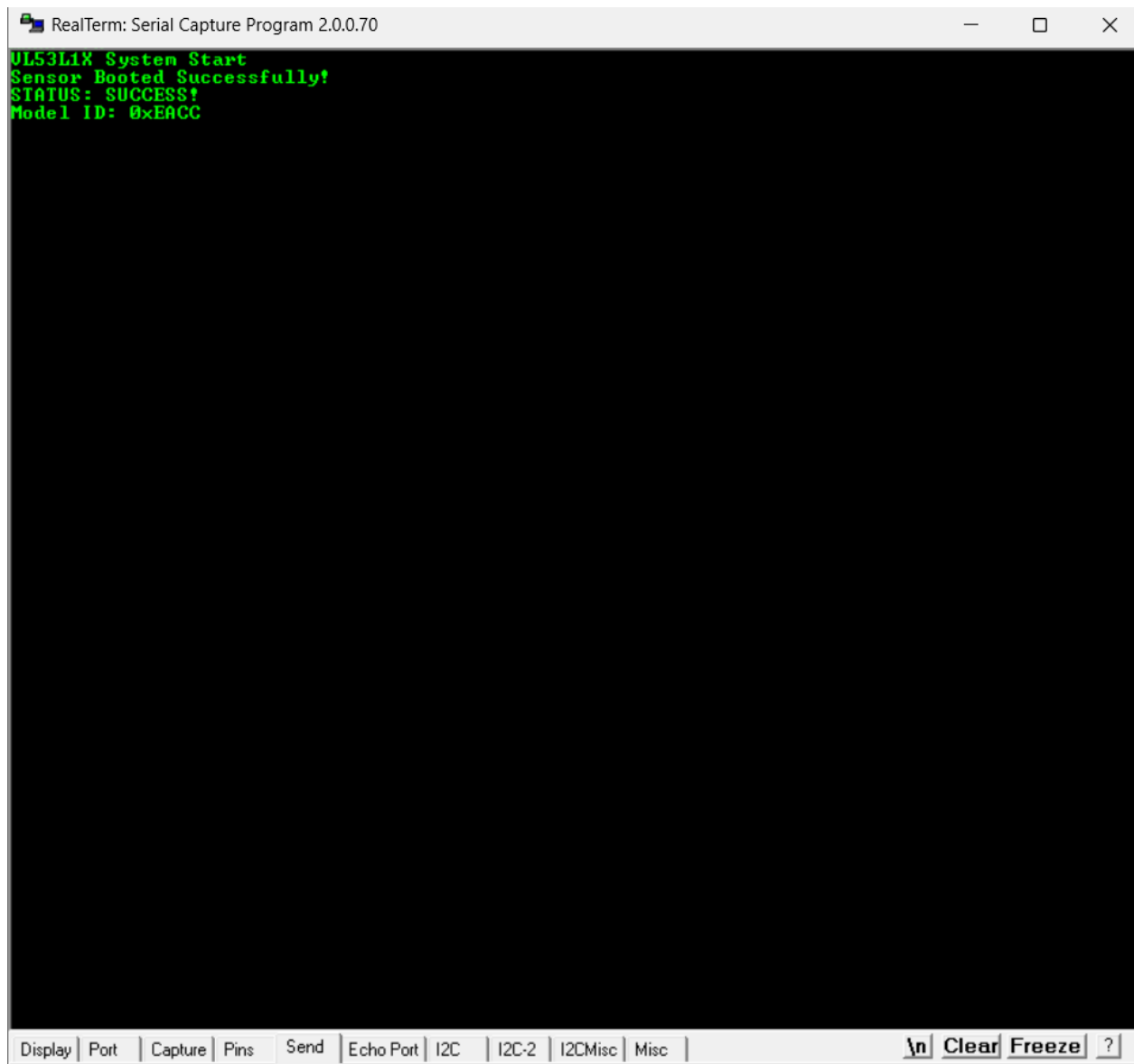


Figure 2: RealTerm Display

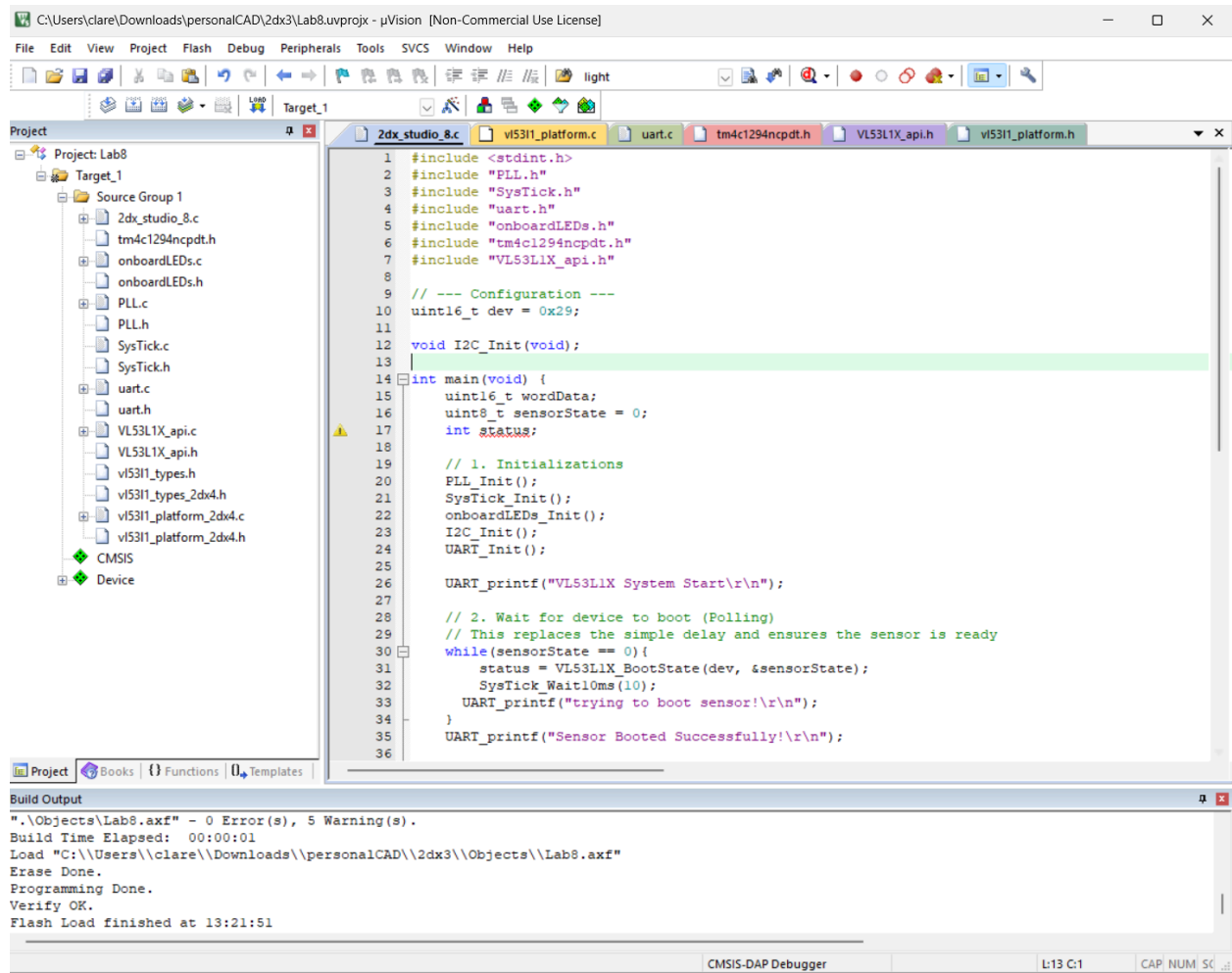


Figure 3: Correct file implementation

Debugging exemplar

The debugging exemplar from my lab work in relation to the theme - Act - is Lab 7 Milestone 1. In this milestone, logical debugging was done on simple arithmetic for the correct time delay of the interrupt. This required solid and basic understanding on how an interrupt works, and how clock cycles function. This milestone shows how important timing is in microsystems, as precise

and fast signalling is balanced according to user needs - implemented through thorough knowledge and understanding by an engineer.

A specific debugging example that is different from the previous theme reports - Observe and Reason - would be the validation of user specifications. This milestone required a 250 ms interrupt service routine. Although the correct waveform was displayed (shown in *Figure 4*), the code implementation did not meet the specifications, and instead produced a rectangular wave with the same 'wait' timing in the code.

The reason why this happened was because `TIMER0_TAILR_R` was set to $120000000 - 1$ instead of $30000000 - 1$. This value was calculated by taking 120MHz (MCU clock) and dividing it by 4 to get 30000000. This value is then subbed into the formula and subtracted by 1 (subtracting by 1 because zero is accounted for as one 'count'). If we instead did $120000000 - 1$, this would mean the timer would wait 1 second.

Therefore, through some trial and errors (as well as some more logical reasoning behind the calculations), the correct waveform with the correct time delay is shown in *Figure 4*, demonstrated using the AD3.

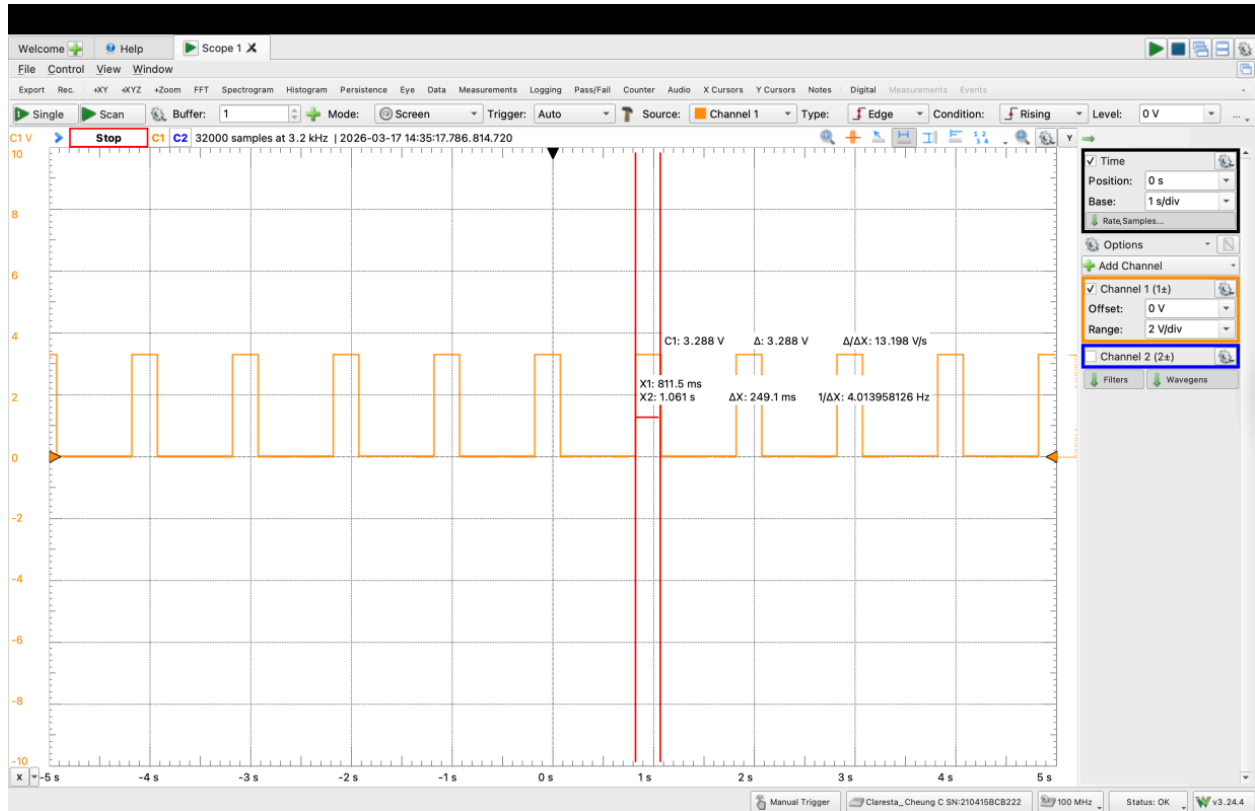


Figure 4: Correct waveform

Synthesis

Across Labs 7 to 8, all three selected theme examples collectively illustrate the theme of Acting in the context of intelligent systems. The microcontroller consistently performs the same fundamental task - performing an action after observation and thorough reasoning of its surrounding environment. For each action it takes, the microcontroller must undergo the aforementioned process to execute an output.

Although each exercise involved different goals to be achieved through the use of interrupts and the ToF sensor, the underlying mechanism remained the same: do something when a specific condition arises after observation and reasoning (if that condition has been met or not). Put

simply, it is an if-then statement. First, observe the factors/variables in the 'code'. If a specific scenario is intercepted (for example, if the stepper motor has turned 45 deg), then the corresponding action is taken (for example, measuring the distance with the ToF sensor after verifying the stepper motor has turned 45 deg).

As a whole, these examples enable intelligent systems to make their observations and reasoning come to life and actually make an impact - most hopefully beneficial and as efficient as the engineer can make it to be.

Reflection

These examples contributed to my understanding of this theme of Acting as I went through the exercises in Labs 7 and 8. These labs provided me with the knowledge and understanding of how a system executes after the topics of the first two theme reports (Observe and Reason). By actually being able to physically see the system work and do something in real life allowed me to further my knowledge on how a microsystem acts.

Lab 7 provided me with the fundamental understanding of how communication between 2 devices work and is executed (acted), specifically between ToF and the MCU. This was a major contribution in introducing Acting in microsystems, allowing easy understanding through simple code integration.

Lab 8 provided me with more fulfilling and useful actions the intelligent system could perform. Built off from Lab 7, it required communication between both devices. The most interesting part

of this Lab was seeing it come together. By using the stepper motor knowledge from previous labs, all themes came together as one. The ToF sensor felt like the 'eyes' of the microsystem, making it surreal. When it has finished observing one area of the space, it reasons whether or not data was collected and received. If so, it continues turning (the stepper motor) and recording the distance measured. This was also extremely useful in terms of furthering our knowledge for Project 2 Deliverable.

References

[1] Y. M. Haddara, *2025–2026 COMPENG 2DX3 Laboratory Manual*, Jan. 2024. [Online].

Available: <https://avenue.cllmcmaster.ca/d21/1e/content/731113/viewContent/5518323/View>.

Accessed: Mar. 25, 2026.

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